TCP/IP and Routing

A network moves data as packets. The TCP protocol provides reliable delivery between hosts, while UDP is faster but does not guarantee that a packet arrives.

Each device has an IP address, and a subnet groups addresses on one network. A router forwards each packet towards its destination using a routing table, and a switch connects devices on a LAN.

DNS translates names into an IP address, and HTTP runs on top of TCP for the web. Ethernet and wireless links carry the packets across the physical network.